Arrival

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Look at the cell and allele models. Point to one detail you notice.

Point, sign, speak, draw, type or write. An adult may read or scribe your exact words. You may pass or take an agreed pause. Plan in any language.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

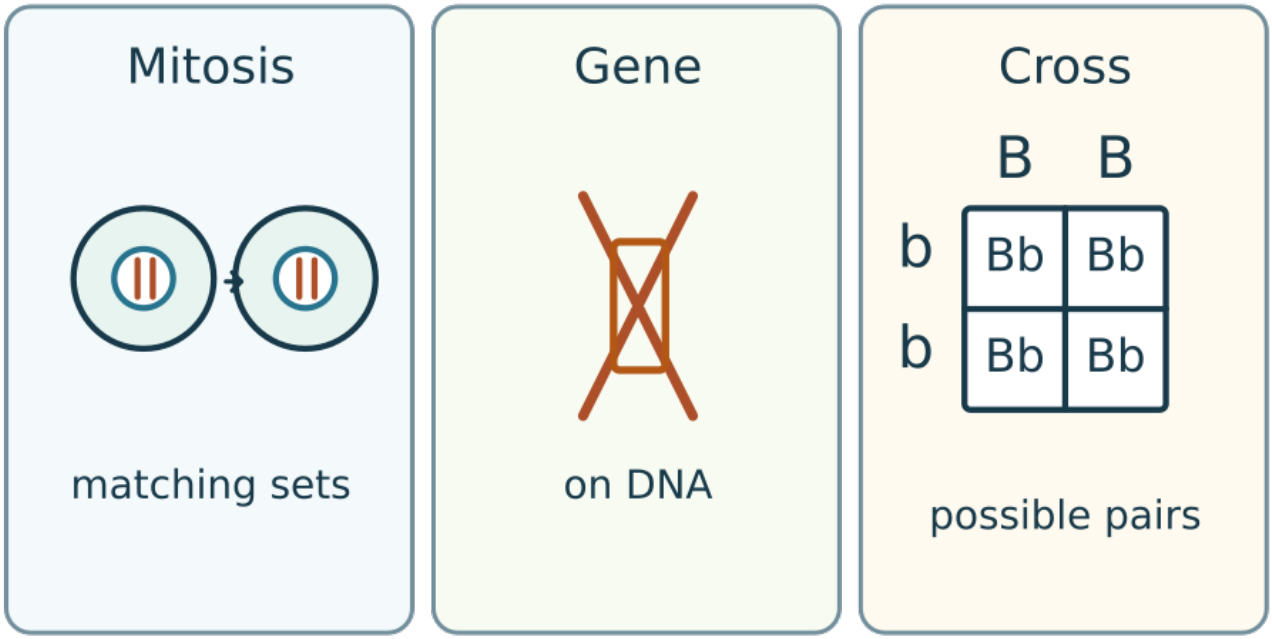
Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Three rehearsal models

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.



These are separate model pictures. The rehearsal cross is BB × bb. Put these pictures away before your own check. Sizes do not match real life.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Starter

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Show one answer at a time. Keep your first try.

1. Which named source supports our cystic fibrosis card?
2. How does a gene relate to DNA?
3. Which part of an animal cell holds chromosomes?

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Word page

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Word	Meaning	Example
mitosis	division of the nucleus to separate matching chromosome sets	Mitosis helps make matching body cells.
gene	a section of DNA that codes for a protein	A gene is a part of the DNA string.
allele	a version of a gene	A and a stand for two alleles in our invented model.
evidence	a fact or model detail that supports a claim	The labelled chromosome sets are evidence for my answer.

Use your own language to plan or explain these words.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

I do · a parallel model

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Label a body-cell model with copied chromosome sets. Explain that mitosis separates them into two nuclei. Point to a detail that supports your answer. Cover the labels before your own check.

Show me: does describe need a model detail or a personal story?

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · the missing step

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

The chromosome sets are copied. Mitosis moves the sets to ___ nuclei. The detail I used is ___.

Supported: use a different model and the command words first.

Independent: explain which model detail supports a claim.

Aspire: give a limit. A diagram shows order, but leaves out some steps.

Choose first on your own. One partner explains and the other asks for the clue. Swap roles. Each learner keeps their own response.

Take an agreed reset within this stage.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · match the evidence

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Card	Your match
Two matching chromosome sets	_____
Short marked part of a DNA string	_____
Four possible allele pairs	_____

Groups: Cell model, DNA model, Cross model. Give a reason for one match.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do 2 · choose a statement

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Which paired statement is scientifically accurate?

A. Mitosis halves chromosome number; a gene is the whole genome.

B. Mitosis removes DNA; an allele is a separate cell.

C. Mitosis makes one unsplit cell; DNA is only in proteins.

D. Mitosis gives matching body cells; a gene is a section of DNA.

Choose privately. Show your answer to the adult. Then give your reason.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Supported

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa × aa to state a possible pair and chance.

Order the three stages and mark one gene on the model. Then use the cross.

First ___; then ___; last ___.

Use the blank check sheet. Put the rehearsal model away. Work on your own for the full 8 minutes. Keep your first response.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Independent

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa × aa to state a possible pair and chance.

Explain matching body cells and why a gene is only part of DNA.

The model shows ___ because ___.

Use the blank check sheet. Put the rehearsal model away. Work on your own for the full 8 minutes. Keep your first response.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Aspire

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa × aa to state a possible pair and chance.

Use Aa × aa to reject the claim that a probability promises an exact family count.

The model chance is ___; the real count may differ because ___.

Use the blank check sheet. Put the rehearsal model away. Work on your own for the full 8 minutes. Keep your first response.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

My own check · cells and DNA

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

1. Put these stages in order: Split / Copy / Separate. Explain what each step does.

My order and reason: _____

2. Mark one short part of the DNA string as a gene in this simple model. Explain your mark.

DNA string model
A T C G G T A T C G

This short string is a model, not a real gene sequence.

My explanation: _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

My own check · a chance

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

3. Use Aa × aa. Label the top and side, fill each box, then give one possible pair and its chance.

Side / top	_____	_____
_____	_____	_____
_____	_____	_____

My possible pair and chance: _____

Does this promise an exact family count? My reason: _____

This is an invented one-gene model. Keep personal and family details private.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

After the check · two claims

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

After you hand in your first response, judge these claims.

Claim A: two aa boxes out of four promise exactly two aa children in every family of four.

Claim B: if two offspring are not aa, the next one must be aa.

Name each error. Repair it with a reason. Keep this work separate from the checked answer.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Review

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Keep your first answer. Hear the actual feedback. Make one real change, or give a reason to keep it. Record the next step you chose.

Mark a true self-edit // and ask for a read-back ? if it helps.

Actual feedback: _____

My change, or why I kept it: _____

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

Show one correct model detail and explain how it supports your answer.

My feedback was: _____

Next I need to: _____

Choose one model error to repair and explain in the next lesson.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Knowledge organiser

I can order copy, separate and split for mitosis.

I can mark one gene as part of DNA.

I can use a simple cross as probability, not a promise.

I am learning to use cell and inheritance models to show what I know and choose one next step.

DNA is copied before mitosis. The copies separate into matching sets in two nuclei. The cell then splits.

A chromosome contains DNA. A gene is one section of DNA. An allele is a version of a gene.

A simple cross shows possible allele pairs and their chances. It does not promise an exact family count.

Use a model detail as evidence. State one limit. Keep your first answer and one precise next step.

Made from the web lesson SCI-LAUNCH-A2-W07
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale